

Unix Programming - Assignment 03

Todo App: Flask + PostgreSQL

CS515 - Unix Programming

IIIT Tiruchirappalli

Dr. Anoop Jacob Thomas

Submitted by: Tanvi Kandoi

GitHub: kandoitanvi27

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1. GitHub Repository

<https://github.com/kandoitanvi27/unix-assignment-3>

The repository contains all source code, Dockerfile, docker-compose.yml, and Kubernetes manifest files.

2. Project Overview

This project is a simple Todo List web application built as a two-tier architecture:

- Frontend/API Layer: A Flask (Python) web application that provides a clean, responsive interface for managing todo items. Users can create, toggle (mark as complete/incomplete), and delete todos.
- Database Layer: PostgreSQL is used as the backend database to persist todo items with fields for id, title, and completion status.

The application demonstrates the use of Git for version control, Docker for containerization, and Kubernetes (Minikube) for orchestration.

3. Step-by-Step Instructions

3.1 Clone the Repository

```
git clone https://github.com/kandoitanvi27/unix-assignment-3.git
cd unix-assignment-3
```

3.2 Run with Docker Compose

```
# Build and start the application
docker compose up --build

# Open in browser: http://localhost:5000
```

3.3 Deploy to Kubernetes (Minikube)

```
# Start Minikube
minikube start --driver=docker

# Build Docker image inside Minikube
eval $(minikube docker-env)
docker build -t todo-app:latest .

# Apply Kubernetes manifests
kubectl apply -f k8s/

# Check that pods are running
kubectl get pods

# Access the application
minikube service todo-app
```

3.4 Kubernetes Manifests Overview

- k8s/db-deployment.yaml: PostgreSQL deployment (1 replica)
- k8s/db-service.yaml: ClusterIP service for PostgreSQL
- k8s/app-deployment.yaml: Todo app deployment (2 replicas)
- k8s/app-service.yaml: NodePort service exposing app on port 30080

4. Proof of Execution

4.1 Git Commit History (git log --oneline)

```
16147b0 fix: add database connection retry on startup
81c4c87 docs: update README with run instructions and add assignment PDF
fadaeec feat: add Kubernetes manifests
2b7a715 feat: add docker-compose for local development
bc96932 feat: add multi-stage Dockerfile
ddccd73 feat: add frontend styling
9a9d150 feat: add CRUD operations for todos
040f68c feat: add database connection and todo table schema
8af5acf feat: add Flask app with basic routes and template
949c2d6 feat: initial project structure
```

4.2 Kubernetes Pods (kubectl get pods)

NAME	READY	STATUS	RESTARTS	AGE
postgres-5b48bb7b7d-5d52r	1/1	Running	0	5m20s
todo-app-65df8cbf6-278d7	1/1	Running	0	2m13s
todo-app-65df8cbf6-58nfc	1/1	Running	0	2m13s

4.3 Application Running in Browser

My Todo List

Add

No todos yet. Add one above!